R_X	R_Y	$d = R_X - R_Y$	d^2
1	1	0	0
2	4	-2	4
3	2	1	1
4	5	-1	1
5	3	2	4
6	9	-3	9
7	7	0	0
8	10	-2	4
9	6	3	9
10	8	2	4
			$\sum d^2 = 36$

Table 9.4

The rank correlation is given by

$$\rho = 1 - \frac{6\Sigma d^2}{N(N^2-1)} = 1 - \frac{6(36)}{10(10^2-1)}$$

$= 1 - 0.218$

$\therefore \rho = 0.782$

Example 9.7

Ten competitors in a beauty contest are ranked by three judges in the following order.

First judge	1	4	6	3	2	9	7	8	10	5
Second judge	2	6	5	4	7	10	9	3	8	1
Third judge	3	7	4	5	10	8	9	2	6	1

Use the method of rank correlation coefficient to determine which pair of judges has the nearest approach to common taste in beauty?

Solution:

Let R_X, R_Y, R_Z denote the ranks by First judge, Second judge and third judge respectively

R_X	R_Y	R_Z	$d_{XY} = R_X - R_Y$	$d_{YZ} = R_Y - R_Z$	$d_{ZX} = R_Z - R_X$
1	2	3	-1	-1	2
4	6	7	-2	-1	3
6	5	4	1	1	-2
3	4	5	-1	-1	2
2	7	10	-5	-3	8
9	10	8	-1	2	-1
7	9	9	-2	0	2
8	3	2	5	1	-6
10	8	6	2	2	-4
5	1	1	4	0	-4

d^2_{XY}	d^2_{YZ}	d^2_{ZX}
1	1	4
4	1	9
1	1	4
1	1	4
25	9	64
1	4	1
4	0	4
25	1	36
4	4	16
16	0	16
$\Sigma d^2_{XY} = 82$	$\Sigma d^2_{YZ} = 22$	$\Sigma d^2_{ZX} = 158$

Table 9.5

$$\rho_{XY} = 1 - \frac{6\Sigma d^2_{XY}}{N(N^2-1)} = 1 - \frac{6(82)}{10(10^2-1)}$$

$= 1 - 0.4969 = 0.5031$

$$\rho_{YZ} = 1 - \frac{6\Sigma d^2_{YZ}}{N(N^2-1)} = 1 - \frac{6(22)}{10(10^2-1)}$$

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